Ahmad Abdal Qader

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Education

Georgia Institute of Technology & Emory University — *Atlanta, GA*
Ph.D., Biomedical Engineering — *Aug 2022 – Dec 2027 (expected)*
Computational Neural Engineering. Advised by Chethan Pandarinath and Ellen J. Hess.
T32 Computational Neural Engineering Training Program Scholar

University of Rochester — *Rochester, NY*
B.S., Biomedical Engineering — Signals and Systems — *Aug 2018 – May 2022*
Cum Laude, General and Departmental Honors

Publications

[1] Y. H. Ali*, **A. Abdal Qader***, P. H. Bechefsky, B. A. Richards, A. Acosta, B. Lacayo, Z. M. Williams, L. R. Hochberg, N. AuYong, C. Pandarinath. Discrete spatial navigation as a complement to point-and-click in brain–computer interfaces. *Journal of Neural Engineering*, in revision. *equal contribution.

[2] **A. Abdal Qader**, Y. Donsante, J. E. Markowitz, H. A. Jinnah, C. Pandarinath, E. J. Hess. Behavioral signature of trihexyphenidyl in the *TOR1A* (DYT1) knockin mouse model of dystonia. *Dystonia* 4:15034, 2025. preprint

Conference abstracts

[3] **A. Abdal Qader**, Y. Donsante, J. E. Markowitz, H. A. Jinnah, C. Pandarinath, E. J. Hess. Behavioral signature of trihexyphenidyl in the *TOR1A* (DYT1) knockin mouse model of dystonia. *Interface Neuro*, Atlanta, GA, May 2025. Poster.

Research Experience

Emory University & Georgia Institute of Technology — *Atlanta, GA*
Graduate Research Assistant — Systems Neural Engineering Lab — *Aug 2022 – Present*

Sample-efficient neural handwriting decoding and cross-modal transfer | *PyTorch, sequence-to-sequence modeling, LoRA*

- Collected an attempted-handwriting dataset of 256-channel intracortical recordings from a BrainGate2 clinical trial participant with tetraplegia; trained and deployed sequence decoders reaching ∼6% offline character error rate and 14% online.
- Trained a cross-modal adapter into Meta's open-source surface EMG handwriting decoder, reusing large-scale non-invasive pretraining for intracortical decoding. Up to 4× more data-efficient than a tuned GRU trained from scratch, and generalized to characters absent from the neural training set.

Real-time surface EMG acquisition and decoding in tetraplegia | *Redis, BRAND, real-time processing*

- Achieved closed-loop one-dimensional cursor control from residual, low-SNR surface EMG in a participant with tetraplegia and little to no functional hand movement, via an online decoding pipeline for a 95-channel high-density wearable band.

- Integrated an additional wearable wristband recording sEMG, PPG, IMU and EDA, synchronizing all streams to simultaneous intracortical signals and laying the groundwork for a multimodal dataset relating neural activity to user state.

Hybrid discrete–continuous BCI-driven on-screen navigation | *closed-loop brain–computer interfaces*

- Integrated velocity decoding and gesture classification into a hybrid navigation paradigm with on-demand switching between fast cursor reaches and precise gesture-based selection, combining the speed of one with the accuracy of the other.
- Trained gesture decoders on accumulated closed-loop data that generalized zero-shot to later sessions, staying accurate over weeks despite distribution shift in the input intracortical signals.

Unsupervised modeling of behavior and neural activity in dystonia models | *computer vision, probabilistic graphical models*

- Quantified how an anti-dystonia drug reorganizes spontaneous mouse behavior by modeling 3D depth video as sequences of sub-second motifs with a hierarchical autoregressive HMM (MoSeq), uncovering consistent drug effects across genotypes and sex differences. Led data acquisition and analysis; first-author paper.
- Extracted single-neuron activity from low-SNR one-photon calcium imaging of the striatum with constrained non-negative matrix factorization (CNMF-E), including motion correction, parameter tuning and batch processing across sessions, plus a configurable simulator for controlled testing of the pipeline.

University of Rochester *Rochester, NY*
Research Assistant — Padmanabhan Lab *May 2021 – May 2022*

- Implemented dynamic segmentation algorithms to extract spatially varying transient signals from fluorescence microscopy of morphing cells, and built in-house acquisition and analysis tooling around them. Cultured human stem cell lines, performed viral transfections and ran imaging experiments.

Research Assistant — Henry Lab *Jun 2020 – Aug 2020*

- Built analog signal-conditioning circuits and 3D-printed components to make reward delivery in a behavioral conditioning apparatus measurable and accurate; ran the behavioral conditioning experiments on avian models of hidden hearing loss.

Conferences and Meetings

Society for Neuroscience (SfN)	*Washington, DC / Nov 2026*
BrainGate Annual Summit	*Santa Cruz, CA / Aug 2026*
BrainGate Annual Summit	*Olympic Valley, CA / Aug 2025*
Interface Neuro Annual Meeting	*Atlanta, GA / May 2025*
BrainGate Annual Summit	*Atlanta, GA / Aug 2023*
Simons Foundation Workshop on Calcium and Voltage Imaging Analysis with CaImAn	*New York, NY / Feb 2023*
NSF Collaborative Research in Computational Neuroscience (CRCNS)	*Atlanta, GA / Oct 2022*
Simons–Emory International Consortium on Motor Control	*Atlanta, GA / Oct 2022*

Teaching and Mentoring

Teaching Assistant — Introduction to Biomechanics, Georgia Tech *Fall 2023, Spring 2024*

- Led the Problem Solving Studio section for 50 undergraduates working in groups of four to five, introducing open-ended biomechanics problems and running solution reviews.

- Designed grading rubrics for exams and quizzes, and supervised the undergraduate teaching assistants responsible for grading.

Undergraduate research mentorship *2023 – Present*

- Mentored three undergraduate researchers, including a SURFiN summer fellow, on data acquisition, analysis pipelines and presentation of results.

Awards and Fellowships

Computational Neural Engineering Training Program Scholar (T32) *Aug 2022*
Cum Laude, General and Departmental Honors, University of Rochester *May 2022*
Davis Scholar — $80,000 undergraduate tuition award *Aug 2018*
International Baccalaureate Merit Scholarship — $40,000 *Aug 2018*
Aurora 100Lives Gratitude Scholarship — $70,000, United World College Dilijan *May 2016*

Technical Skills

Concepts: deep learning, time-series sequence modeling, transfer learning and domain adaptation, unsupervised learning, probabilistic graphical models, real-time signal processing, experimental design

Architectures: RNNs, CNNs, transformers, state-space models, autoregressive HMMs

Modalities: time series (intracortical signals, high-density sEMG, IMU, PPG, EDA); video (3D depth, one-photon calcium imaging)

Packages and tools: PyTorch, NumPy, SciPy, scikit-learn, Matplotlib, Weights & Biases, Redis, BRAND, DeepLabCut, CaImAn

Languages: Python, MATLAB, bash, LaTeX

Hardware: real-time data acquisition, multi-electrode arrays, circuit design, microcontrollers

Other Experience

AS&E Information Technology Center, University of Rochester *Aug 2018 – May 2022*

- **IT Lead and Training Coordinator** (2019–2022) — supervised and trained a team of student IT workers, built modular training materials, managed schedules and coverage, and planned the transition to remote operations in 2020. Previously Operations and End-User Support Consultant (2018–2019).